# EvidEO: ask a question about land, get an answer you can check.

Free satellites photograph all of India every few days. EvidEO lets anyone ask a plain question about a place and a date, and answers with a measurement taken from those images, the proof behind it, and an honest "not enough evidence" when it cannot be sure.

## The problem

Satellite images could answer urgent questions like "which villages are flooded?" or "did this crop fail?". Today, using them is hard:

- **It needs specialist skills.** Choosing satellites, downloading huge files and processing them takes mapping (GIS) training most offices do not have.
- **Clouds hide the ground** in the monsoon, exactly when floods happen. Ordinary photos from space show only cloud.
- **Official maps are shared thin.** In 2024 India's national flood-mapping service issued about 300 flood maps for the whole country, so most local questions go unanswered.
- **AI chatbots sound sure but show no proof.** A fluent answer with no date, image or confidence is hard to act on.

## Who it helps

**District disaster officials** "Which parts of my district are under water since Tuesday, and is it spreading?"

**Agriculture and crop-insurance officers** "Did crops in these blocks really suffer this season?"

**Journalists and fact-checkers** "Was this place really flooded on the date a viral post claims?"

**Planners and researchers** "How much has this town grown since 2020?"

We start with **monsoon floods and crop condition at district level**, where a slow or wrong answer costs the most.

## How it works, in four steps

1. **You ask**
   Type a question in plain words. EvidEO repeats back what it understood, so a misreading is caught at once.
2. **It finds the right image**
   It searches free public satellite archives. If clouds block the view, it switches to radar, which sees through cloud.
3. **It measures**
   A transparent, well-tested method does the measuring (for example, spotting water in radar images). The AI never makes up the number.
4. **It shows the proof**
   You get an evidence card: the answer, a map, which image and date it came from, and how confident it is.

### WHAT AN ANSWER LOOKS LIKE

**"How much of Barpeta district was under water around 11 July 2024?"**



**About ___ sq km under water**

Confidence: high · Radar image, 11 Jul 2024 · Scene ID shown · Re-run receipt

*Illustration of the layout only. The measuring engine is being built now, so no real figure is shown.*

## And when it is not sure

If the image is unclear, the question is outside what EvidEO can measure, or confidence falls below a set level, it says **"insufficient evidence"** instead of guessing, and tells you what would help, such as the date of the next satellite pass.

A careful "I don't know" protects decisions. A confident wrong answer about a flood or a crop can send help to the wrong place or deny a fair claim.

## Why it is easier than what exists

Our innovation is not a bigger AI model. It is a shorter, checkable path from a question to an answer someone can act on.

| | TYPICAL SATELLITE AI CHATBOT OR GIS TOOL | EVIDEO |
|---|---|---|
| **Skills needed** | GIS training, or knowing which satellite and settings to use | **Just a question in plain words** |
| **Monsoon clouds** | Often no usable image, or you must know to switch to radar | **Switches to radar automatically and says so** |
| **Where the number comes from** | Generated by the AI model itself | **Measured by a transparent method from the actual image** |
| **Proof** | Usually none shown | **Image ID, date, map, confidence and a receipt anyone can re-run** |
| **When unsure** | Usually answers anyway | **Says "insufficient evidence" and what would fix it** |
| **Cost and setup** | Paid licences, powerful hardware, or region-locked services | **Free public data, open models, runs on an ordinary computer, can run offline** |

### Why you can trust it

- **Every answer points to its source:** the exact satellite image, its date and the area measured.
- **Confidence is checked, not guessed.** It is tuned against cases where the true answer is known, and shown in plain words: high, moderate or low.
- **Anyone can re-run it.** Each answer has a receipt that reproduces the same result.
- **Evidence first, chat second.** Studies show AI explanations can make people accept wrong answers as readily as right ones, while a well-tuned confidence helps them judge when to rely on it. So EvidEO leads with the map and proof.

### How it grows

- **More users:** big areas are split into small tiles worked on in parallel; repeated questions reuse results.
- **More places:** it reads only the part of each image it needs, with backup archives if one is down.
- **More languages:** only the question reading changes; numbers never pass through translation.
- **Government use:** runs on a laptop, a state data centre, or fully offline.

### What it will not do

Count cars or houses (the free images are too coarse), predict the future, or replace official maps and field checks. It answers the local follow-up questions those cannot.

### Where the project is today

Done Research review: 75 papers so far, of about 200 planned

Done System design and working app skeleton

Done Connection to free satellite archives, with automatic radar fallback

Next Measuring engine: floods, crop change, land cover

Then Full app screens, training code, testing, demo

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### Words used here

**Radar (Sentinel-1):** a European satellite that sends its own signal, so it sees through cloud and at night.

**Optical (Sentinel-2):** a satellite camera, like a photo from space; blocked by cloud.

**GIS:** mapping software that usually needs training.

**Confidence:** how likely the answer is to be right, checked against known cases.